Computer Networking Basics [week 2]

1. What is Computer Networking?
   - ➔ Computer networking is the practice of connecting computing devices to share data, resources, and services efficiently.
   - ➔ In simple terms, a computer network allows devices to send and receive information efficiently, whether they are in the same building or located anywhere in the world.
   - ➔ Example: When you open a website, your computer send a request over the internet to the web server hosting that website. The server processes the request and sends the webpage back to your computer. This communication happens through a computer network.

2. Types of Networks (LAN, MAN & WAN)
   1. LAN: LAN stands for Local Area Network. It connects computers and devices within a small geographical area, such as schools, campus, home. LANs provide high-speed communication and allow users to share files, printers, and internet connections.
      Example: The network connecting computers in a school computer lab or an office.
   2. MAN: MAN stands for Metropolitan Area Network. It connects multiple LANs across a city or metropolitan area. It is larger than LAN but smaller than WAN and is commonly used by an organization with multiple branches in the same city.
      Example: A university connecting its campuses across a city or a city's cable TV network.
   3. WAN: WAN stands for Wide Area Network. It connects computers and networks over large geographical distance, such as across countries. It enables communication between remote locations using public or private communication links.
      Example: The Internet is the largest and most widely used WAN, connecting millions of devices worldwide.

3. What is an IP Address?
   - ➔ An IP(Internet Protocol) Address is a unique identifier assigned to every device connected to a network. It allows device to identify and communicate with each other over a local network or the internet.
   - ➔ There are two versions of IP Addresses: IPv4 and IPv6.

4. Difference between Public IP and Private IP?
   - ➔ Public IP: Used to identify your network on the Internet. It is globally unique and accessible over the internet. Used for communication between devices on the internet. Assigned by the ISP(Internet Service Provider).
     Example: 49.205.123.45

- ➔ Private IP: Used to identify devices within your local network (LAN). It is not accessible directly from the internet. Used for communication between devices within the same local network. Assigned by a router.
  Example: 192.168.1.10, 10.0.0.5 .

5. What is DNS? What is a Port?
   - ➔ **DNS(Domain Name System)** is the system that translate human-readable domain-names into IP addresses that computer uses to communicate over a network. Without DNS, users would have to remember the IP addresses of every website instead of domain name.
   - ➔ **Port** is a logical endpoint that identifies a specific service on device. Port numbers range from **0 to 65535**. Each service uses a specific port number so that network traffic reaches the correct application.

6. Basic Network Troubleshooting Commands.
   - ping: verifies network connectivity between your computer and another device or server.
   - ip addr/ ifconfig: displays the current IP configuration of your device and helps identify network configuration issues.
   - netstat: shows active connections and listening ports, helping diagnose network and application-related issues.
   - traceroute: helps identify where delays or failures occur along the network path.
   - nslookup: confirms if a domain name is correctly translated into an IP address by the DNS server.